

Mini Project 01

Two-Stage Miller OTA

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PART 1: g_m/I_D Design Charts

In this section, the required design charts are generated using ADT Device Xplore based on the g_m/I_D methodology.

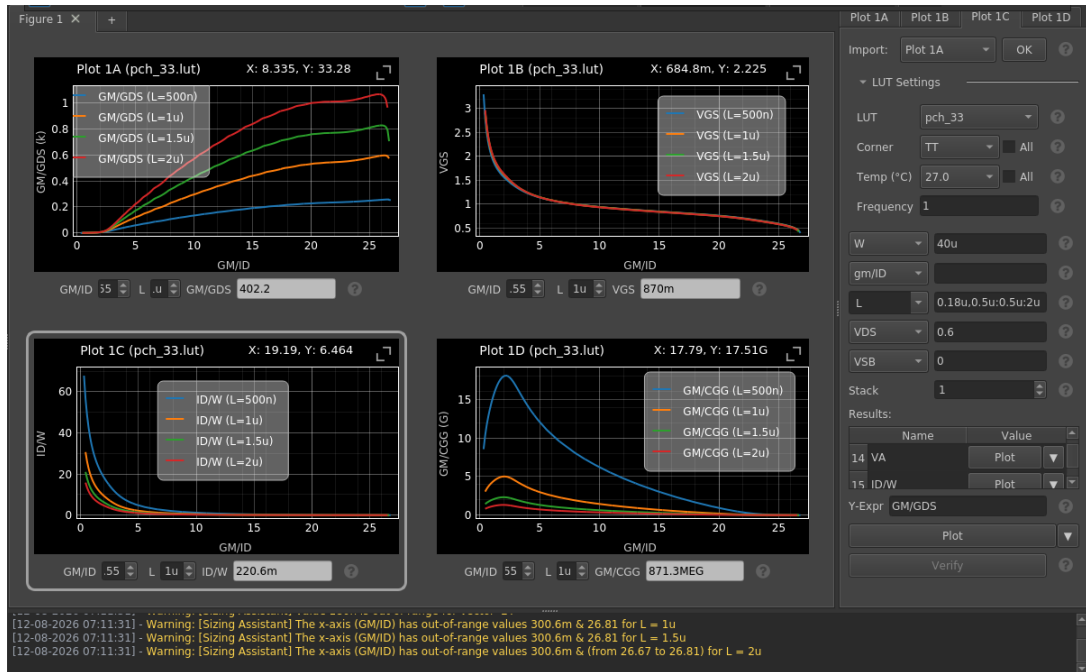


Figure 1: PMOS Design Charts: g_m/g_{ds} , V_{GS} , I_D/W , and g_m/C_{gg} versus g_m/I_D at $V_{DS} = 0.6$ V.

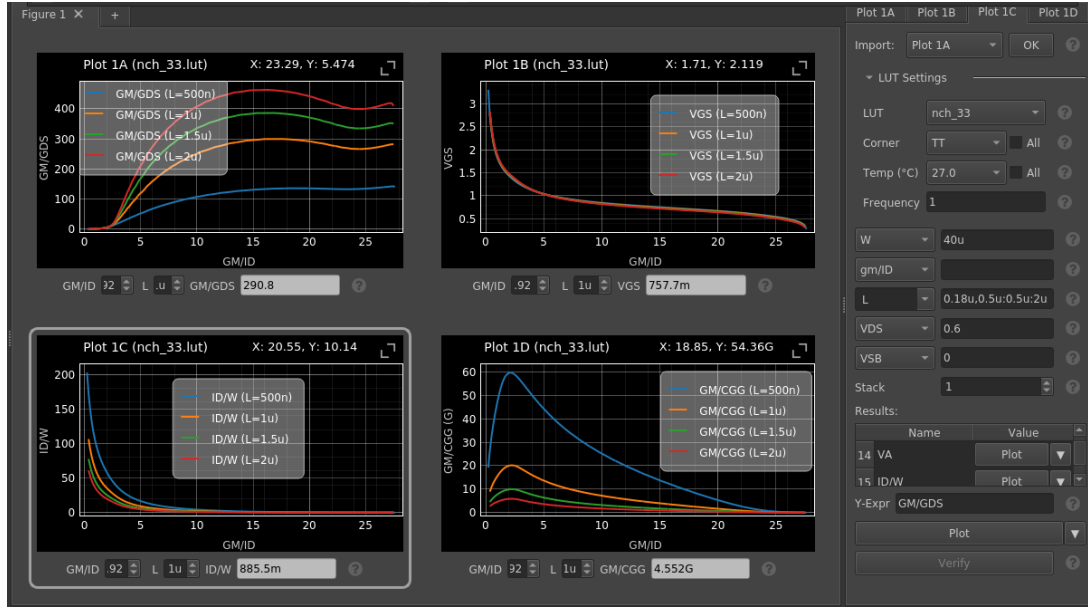


Figure 2: NMOS Design Charts: g_m/g_{ds} , V_{GS} , I_D/W , and g_m/C_{gg} versus g_m/I_D at $V_{DS} = 0.6$ V.

1 PART 2: Design Specifications

1.1 Open-Loop Gain and Static Gain Error

For a unity-gain buffer configuration where the feedback factor is $\beta = 1$, the closed-loop gain error is specified to be $\leq 0.05\%$ (0.0005). The static error in a feedback system is related to the open-loop DC gain (A_{ol}) by:

$$\text{Error} = \frac{1}{1 + \beta A_{ol}} \approx \frac{1}{\beta A_{ol}}$$

Substituting $\beta = 1$ and the maximum allowable error into the equation yields the minimum required open-loop DC gain:

$$A_{ol} \geq \frac{1}{0.0005} = 2000 \text{ V/V}$$

In decibels, this corresponds to:

$$A_{ol, \min} = 20 \log_{10}(2000) \approx 66.02 \text{ dB}$$

1.2 Common-Mode Rejection Ratio (CMRR)

(CMRR = A_{v1}/A_{cm}). :

$$\text{CMRR} \approx A_{ol} \cdot 2g_{m, \text{load}} \cdot R_{SS} (R_{SS} = r_{o3})$$

To satisfy the lab specification of $\text{CMRR} \geq 74 \text{ dB}$, we can calculate the required range/minimum bound for the product of the active load transconductance ($g_{m, \text{load}}$) and

the tail current source output resistance (R_{SS}). Substituting the minimum required differential gain $A_{v1} = 70 \text{ V/V}$:

$$2g_{m,\text{load}}R_{SS} \geq \frac{\text{CMRR}_{\text{req}}}{A_{dm}} = \frac{5012}{70} = 71.6$$

Thus, the product $2g_{m,\text{load}}R_{SS}$ must be at least 71.6 to guarantee that the $\geq 74 \text{ dB}$ CMRR requirement is met.

1.2.1 Minimum Common-Mode Voltage ($V_{ICM,\text{min}}$)

To determine the lower boundary of the common-mode range, we must ensure that the input PMOS transistors remain in the saturation region. The condition for a PMOS device to stay in saturation is that its drain-to-gate voltage (V_{DG}) must not exceed the magnitude of its threshold voltage:

$$V_{DG} \leq |V_{thp}|$$

The saturation condition :

$$V_{GS,\text{nmos}} - V_{ICM} \leq |V_{thp}|$$

$$V_{GS,\text{nmos}} - |V_{thp}| \leq 0.2$$

1.2.2 Maximum Common-Mode Voltage ($V_{ICM,\text{max}}$)

To determine the upper boundary of the common-mode range, we must ensure that the PMOS tail current source (M_3) remains in the saturation region. As the input common-mode voltage (V_{ICM}) increases toward V_{DD} , it pushes the source voltage of the input differential pair upward, so it will compresses the voltage across M_3 and decrease V_{sd} to be less than $v_{sd}(\text{sat})$ so to stay in saturation we should do this condition:

$$V_{SD3} \geq V_{dsat3}$$

the absolute maximum allowable common-mode input voltage:

$$V_{ICM} \leq V_{DD} - V_{SG} - V_{dsat3}$$

To ensure that M_3 remains in the saturation region, design choice is applied by enforcing a minimum voltage drop of $V_{SD3} = 0.3 \text{ V}$.

1.3 Compensation Capacitor and Slew Rate

The Miller compensation capacitor (C_c) is initially sized based on the load capacitance (C_L). the compensation capacitor is determined as:

$$C_c = 0.4 \cdot C_L = 0.4 \cdot 5 \text{ pF} = 2 \text{ pF}$$

With the compensation capacitor defined, the required tail bias current (I_B) can be derived from the Slew Rate (SR) specification. :

$$SR = \frac{I_B}{C_c}$$

The target Slew Rate for the design is $= 5 \text{ V}/\mu\text{s}$. we can solve for the required tail bias current:

$$I_B = SR \cdot C_c$$

$$I_B = (5 \times 10^6 \text{ V/s}) \cdot (2 \times 10^{-12} \text{ F}) = 10 \mu\text{A}$$

A minimum tail current of $I_B = 10 \mu\text{A}$ is required to satisfy the Slew Rate specification while matching our selected compensation capacitance.

1.4 Transconductance Specifications (g_{m1} and g_{m2})

The required transconductance for the differential input pair (g_{m1}) is calculated by the closed-loop transient response, specifically the rise time (t_{rise}). For a single-pole system response, the rise time is related to the time constant (τ) by:

$$t_{rise} = 2.2\tau$$

Given the maximum allowable closed-loop rise time of $t_{rise} = 70 \text{ ns}$, the required time constant is:

$$\tau = \frac{70 \times 10^{-9}}{2.2} \approx 31.82 \text{ ns}$$

The time constant is inversely proportional to the unity-gain frequency (ω_u):

$$\omega_u = \frac{1}{\tau} = \frac{1}{31.82 \times 10^{-9}} \approx 31.43 \times 10^6 \text{ rad/s}$$

For a Miller-compensated OTA:

$$\omega_u \approx \frac{g_{m1}}{C_c}$$

$$g_{m1} = \omega_u \cdot C_c = (31.43 \times 10^6) \cdot (2 \times 10^{-12}) \approx 62.86 \mu\text{A/V}$$

To achieve a Phase Margin (PM) of approximately 76° , the non-dominant pole (ω_{p2}) should be placed at four times the unity-gain frequency:

$$\omega_{p2} \approx 4\omega_u$$

$$\omega_{p2} \approx \frac{g_{m2}}{C_L}$$

Equating the two relationships for ω_{p2} provides a ratio between the two stages:

$$\frac{g_{m2}}{C_L} = 4 \left(\frac{g_{m1}}{C_c} \right)$$

$$g_{m2} = 4 \cdot (62.86 \mu\text{A/V}) \cdot \left(\frac{5 \text{ pF}}{2 \text{ pF}} \right) = 10 \cdot 62.86 \mu\text{A/V} \approx 628.6 \mu\text{A/V}$$

1.5 Input Pair Design and Sizing

The branch current for each transistor in the differential input pair is half of the tail bias current:

$$I_D = \frac{I_B}{2} = 5 \mu\text{A}$$

To provide a design margin, the transconductance is increased to $g_{m1} = 65 \mu\text{A/V}$. Assuming the input PMOS and the active NMOS load have equal output conductances ($g_{ds,\text{in}} = g_{ds,\text{load}} = g_{ds}$), the first-stage DC gain target is set to $A_{v1} = 70 \text{ V/V}$. The required output conductance is calculated as:

$$A_{v1} = \frac{g_{m1}}{r_{o,\text{in}} \parallel r_{o,\text{load}}} = \frac{g_{m1}}{2g_{ds}}$$

$$g_{ds} = \frac{g_{m1}}{2A_{v1}} = \frac{65 \mu\text{A/V}}{2 \cdot 70} \approx 0.46 \mu\text{S}$$

But I choose a $g_{ds} > 460 \text{ nS}$ to reduce the gain, allowing me to increase the second-stage g_m to push the second pole to higher frequencies and increase phase margin. ($I_D = 5 \mu\text{A}$, $g_m = 65 \mu\text{A/V}$, $g_{ds} = 0.46 \mu\text{S}$, and $V_{DS} = 0.4 \text{ V}$), the extracted PMOS device dimensions and threshold voltage are:

- $W = 7.104 \mu\text{m} \approx 7.1 \mu\text{m}$
- $L = 437 \text{ nm} \approx 430 \text{ nm}$
- $V_{thp} = 781.4 \text{ mV}$
- $g_{ds} = 550 \text{ nS}$

This extracted threshold voltage (784.2 mV) will be applied directly to the V_{ICM} specifications.

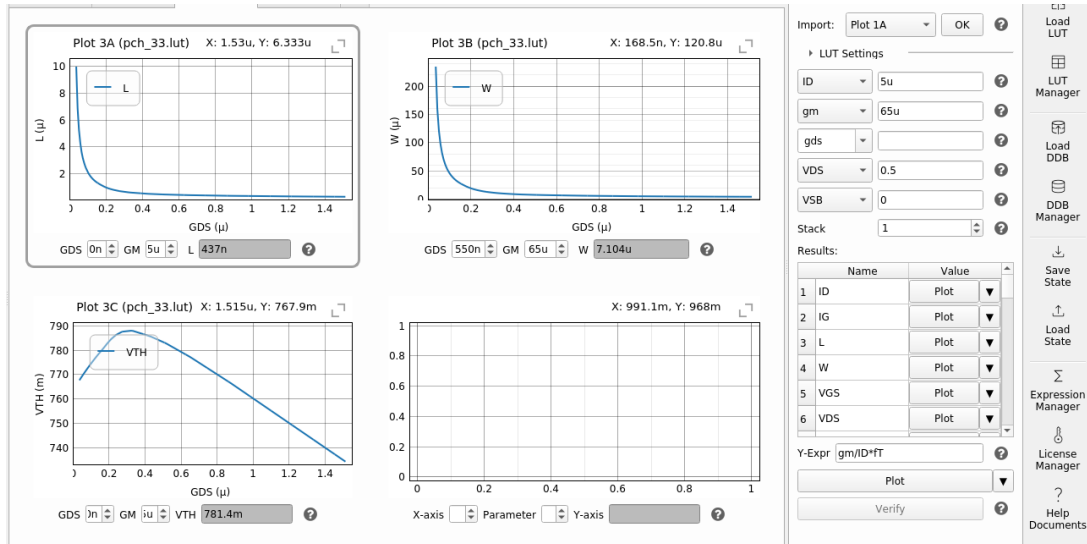


Figure 3: Analog Designer's Toolbox (ADT) parameter extraction for the PMOS input pair.

1.6 NMOS Active Load Sizing

The NMOS active load transistors share the same branch current ($I_D = 5 \mu\text{A}$) and target output conductance ($g_{ds} = 550 \text{ nS}$) as the input pair to satisfy the first-stage DC gain requirement. A nominal drain-to-source voltage of $V_{DS} = 0.5 \text{ V}$ is maintained. Because this load has the same VGS of the NMOS that drive CS amplifier so i will choose the same $gm_2/id_2 = 650 \mu\text{S}/40 \mu\text{A} = 16.25$ so I choose the gm/id of this load =16

the extracted device dimensions are:

- $W = 5.2 \mu\text{m}$
- $L = 560 \text{ nm}$
- $V_{GS} = 734.6 \text{ mV}$

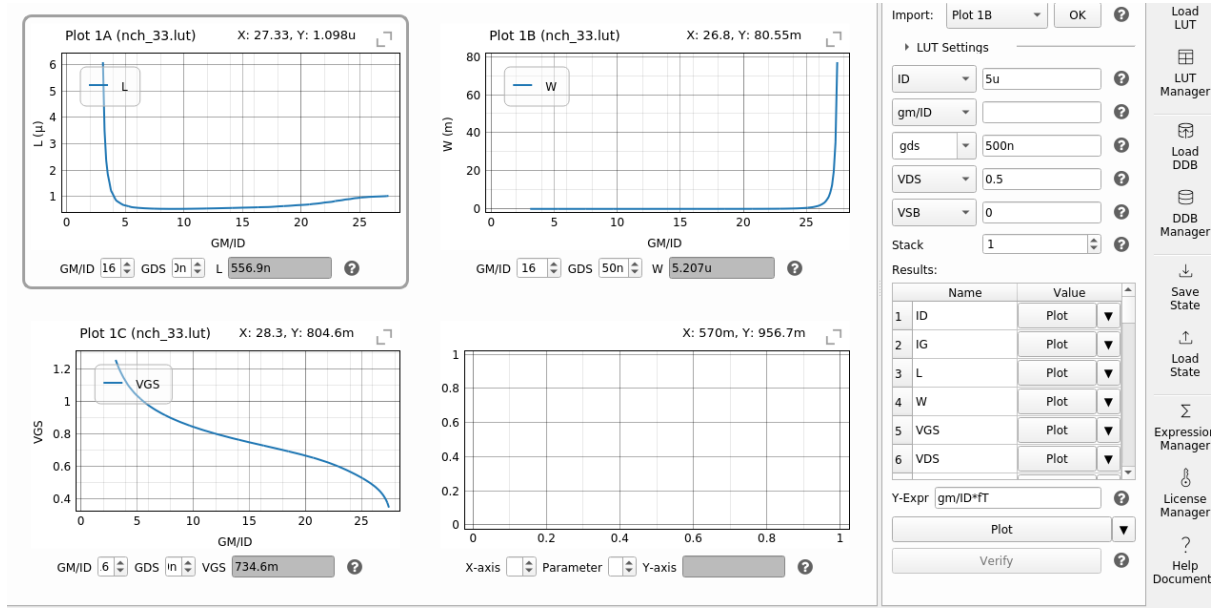


Figure 4: Analog Designer's Toolbox (ADT) parameter extraction for the NMOS active load.

1.7 Second-Stage NMOS Driver Sizing

The NMOS driver for the second-stage Common-Source (CS) amplifier is biased using the same gate-source voltage as the first-stage active load ($V_{GS} = 734 \text{ mV}$). The drain current is scaled up to $I_D = 40 \mu\text{A}$ ($4 \times I_B$).

To maximize the symmetrical output voltage swing and ensure the DC output is centered, the drain-to-source voltage is set exactly at half the supply voltage ($V_{DS} = V_{DD}/2 = 0.9 \text{ V}$). the SECOND-stage DC gain target is set to $A_{v2} = 30 \text{ V/V}$. The required output conductance is calculated as:

$$A_{v2} = \frac{g_{m2}}{r_{o,in} \parallel r_{o,load}} = \frac{g_{m2}}{2g_{ds}}$$

$$g_{ds} = \frac{g_{m2}}{2A_{v2}} = \frac{650 \mu\text{A/V}}{2 \cdot 30} \approx 11 \mu\text{S}$$

Using these operating conditions in the Analog Designer's Toolbox (ADT), the extracted device parameters are:

- $g_m/I_D \approx 13$
- $W \approx 11 \mu\text{m}$
- $L = 320 \text{ nm}$
- $g_{ds} = 12 \mu\text{S}$

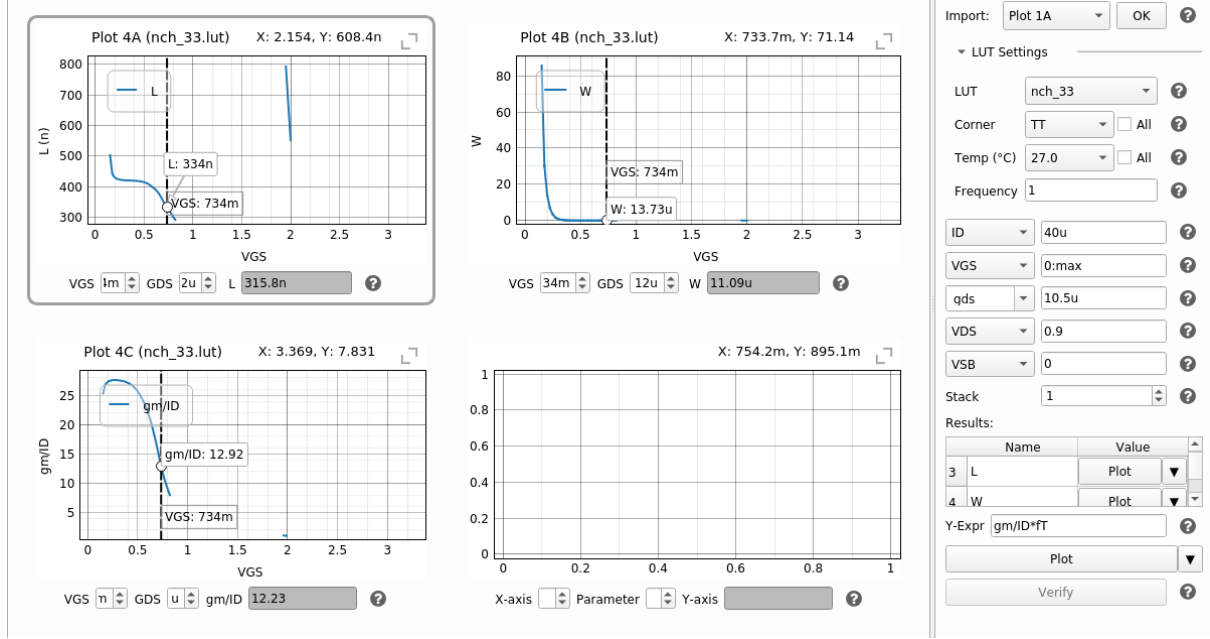


Figure 5: Analog Designer's Toolbox (ADT) parameter extraction for the second-stage NMOS driver.

1.8 Primary PMOS Tail Current Source Sizing

The primary PMOS tail current source for the differential input pair (M_3) provides the total bias current of $I_B = 10 \mu\text{A}$. To guarantee that the device remains in saturation and supports the maximum common-mode input voltage ($V_{ICM,max}$), the drain-to-source voltage is constrained to $V_{DS} = 0.3 \text{ V}$.

$$2g_{m,load}R_{SS} \geq \frac{\text{CMRR}_{\text{req}}}{A_{dm}} = \frac{5012}{70} = 71.6$$

$$R_{ss} \geq 550K$$

Using these operating targets ($I_D = 10 \mu\text{A}$, $V_{DS} = 0.3 \text{ V}$, $R_{ss} = 750K \text{ ohm}$) the extracted PMOS device dimensions are:

- $W \approx 11.2 \mu\text{m}$
- $L \approx 600 \text{ nm}$

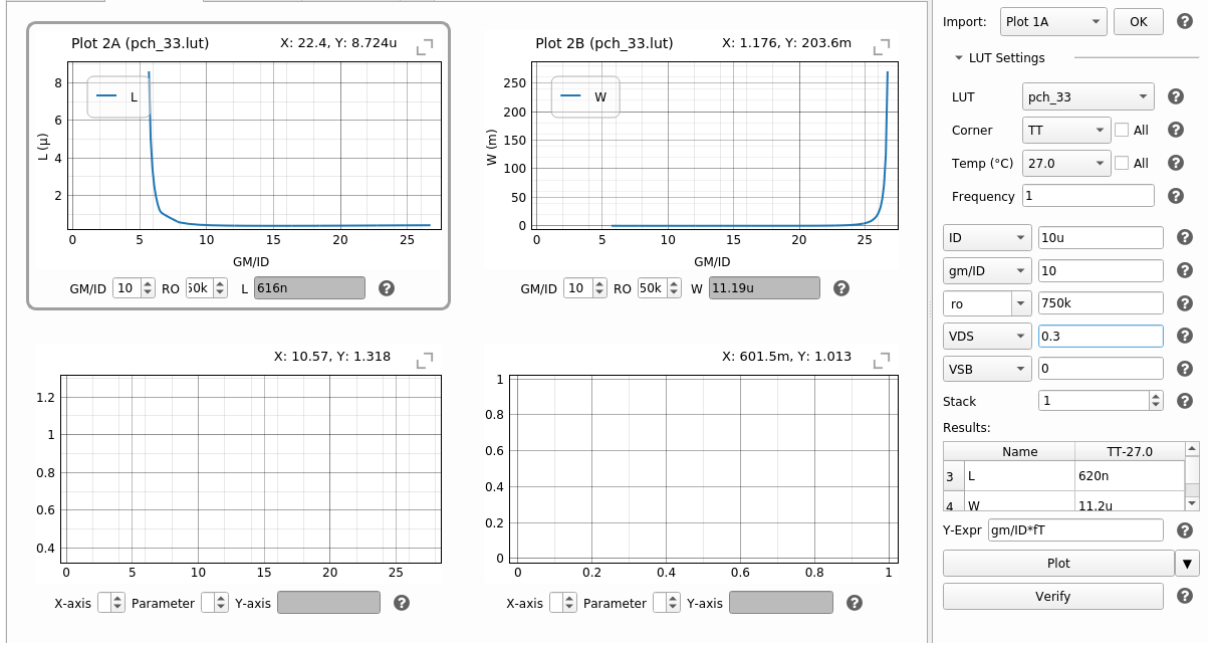


Figure 6: Analog Designer's Toolbox (ADT) parameter extraction for the primary PMOS tail current source.

1.9 Resistor (R_Z)

The RHP zero is located at $\omega_z = g_{m7}/C_c$. To push this zero to infinity, the resistor value is calculated as:

$$R_Z = \frac{1}{g_{m7}}$$

From the DC operating point analysis, the transconductance of the second-stage driver (M_7) is $g_{m7} = 508.3 \mu\text{S}$. Thus:

$$R_Z = \frac{1}{508.3 \mu\text{S}} \approx 1.967 \text{ k}\Omega$$

To ensure the zero is pushed into the left-half-plane and provides a sufficient boost to the phase margin under process variations, I increase this value to $R_Z = 2 \text{ k}\Omega$.

Table 1: ADT VALUES

Device	Type	Function	W	L	g_m	I_D	g_m/I_D	V_{dsat}	V_{ov}	V^*
M3	PMOS	Tail Current Source	$11.2 \mu\text{m}$	600 nm	$98 \mu\text{S}$	$-10 \mu\text{A}$	10	-145.7 mV	-158.1 mV	202.2 mV
M8	PMOS	Inverting Input	$7.1 \mu\text{m}$	430 nm	$64.25 \mu\text{S}$	$-5 \mu\text{A}$	13	-131.8 mV	-97 mV	155.6 mV
M6	NMOS	Active Load	$5.2 \mu\text{m}$	430 nm	$64.25 \mu\text{S}$	$5 \mu\text{A}$	15.67	102 mV	31.4 mV	127.7 mV
M7	NMOS	Stage 2 CS Driver	$11.1 \mu\text{m}$	320 nm	$474.8 \mu\text{S}$	$40 \mu\text{A}$	12.34	124.8 mV	70.4 mV	162 mV

2 PART 3: Open-Loop OTA Simulation

2.1 Schematic and Device Choice Rationale

The top-level schematic of the Miller OTA, including the biasing circuitry, the newly added nulling resistor, and the testbench setup, is presented below.

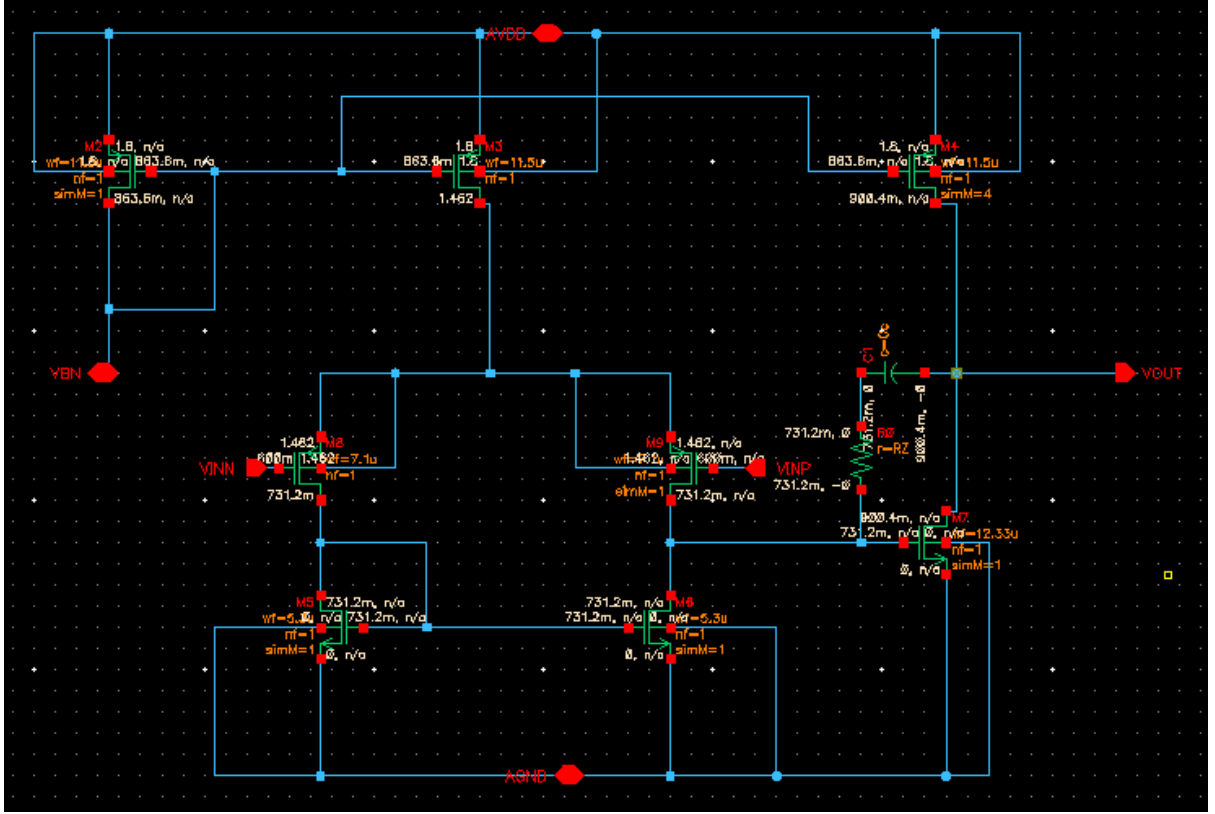


Figure 7: Top-level schematic of the Miller OTA design.

Choice of PMOS vs. NMOS: A PMOS input differential pair (M_8 , M_9) is chosen primarily because the input range near the ground and for its noise performance, as PMOS devices typically exhibit lower flicker ($1/f$) noise compared to NMOS devices of the same area. Furthermore, using a PMOS input pair make the first stage to use an NMOS active current mirror load (M_5 , M_6).the second-stage driver (M_7) as an NMOS transistor becaus in the common-source second stage is effective because its higher electron mobility provides a larger transconductance (g_m) for a given bias current, which pushes the non-dominant pole to a higher frequency and improves overall phase margin and stability.

2.2 DC Operating Point

Table 2: DC Operating Parameters of All MOSFETs

Device	Type	Function	g_m	I_D	g_m/I_D	r_o	Region	V_{dsat}	V_{ov}	V^*
M2	PMOS	Bias Reference	$105.0 \mu S$	$-10.0 \mu A$	10.50	$2.474 M\Omega$	2	$-138.5 mV$	$-146.2 mV$	190.4 mV
M3	PMOS	Tail Current Source	$101.2 \mu S$	$-9.656 \mu A$	10.48	$997.0 k\Omega$	2	$-138.3 mV$	$-146.1 mV$	190.8 mV
M8	PMOS	Inverting Input	$66.08 \mu S$	$-4.828 \mu A$	13.69	$1.894 M\Omega$	2	$-123.6 mV$	$-83.1 mV$	146.1 mV
M9	PMOS	Non-Inverting Input	$66.08 \mu S$	$-4.828 \mu A$	13.69	$1.894 M\Omega$	2	$-123.6 mV$	$-83.1 mV$	146.1 mV
M5	NMOS	Active Load (Ref)	$77.78 \mu S$	$4.828 \mu A$	16.11	$2.328 M\Omega$	2	$98.2 mV$	$24.5 mV$	124.1 mV
M6	NMOS	Active Load (Mirror)	$77.78 \mu S$	$4.828 \mu A$	16.11	$2.328 M\Omega$	2	$98.2 mV$	$24.5 mV$	124.1 mV
M4	PMOS	Stage 2 Active Load	$419.3 \mu S$	$-39.94 \mu A$	10.50	$602.0 k\Omega$	2	$-138.5 mV$	$-146.2 mV$	190.4 mV
M7	NMOS	Stage 2 CS Driver	$508.3 \mu S$	$39.94 \mu A$	12.73	$87.45 k\Omega$	2	$122.2 mV$	$65.8 mV$	157.1 mV

DC Operating Point Analysis

Based on the DC simulation results, the following observations validate the biasing of the OTA:

- **Is the current (and g_m) in the input pair equal?**
Yes, the drain current (I_D) and transconductance (g_m) in the input differential pair (M_8 and M_9) are equal. because there is no mismatch and has the same vds .
- **What is the DC voltage at the output of the first stage? Why?**
The internal DC voltage at the output of the first stage (V_F) rests at approximately 731.8 mV. This voltage is constrained by the supply voltage minus the voltage drops across the primary tail current source and the input PMOS transistor, which can be approximated as $V_F \approx V_{DD} - |V_{DS3}| - |V_{DS8}| = 731.8 \text{ mV}$.
- **What is the DC voltage at the output of the second stage? Why?**
The DC voltage at the output of the second stage (V_{OUT}) is 900 mV. Because the supply voltage is $V_{DD} = 1.8 \text{ V}$, centering the output at $V_{DD}/2 = 0.9 \text{ V}$ is essential to get the maximum possible output voltage swing for the amplifier.

2.3 Differential Small Signal Characteristics

Test	Output	Nominal	Spec	Weight	Pass/Fail
millerr:millerr_tb:2	BW	1.891k			
millerr:millerr_tb:2	Ao_db				
millerr:millerr_tb:2	GBW	5.051M			
millerr:millerr_tb:2	UGF	4.872M			
millerr:millerr_tb:2	Ao	2.671k			
millerr:millerr_tb:2	/VOUT				

Figure 8: Simulated small-signal AC performance metrics.

From the simulation output, the differential parameters are:

- **Open-Loop DC Gain (A_o):** 2.671 kV/V (which corresponds to 68.53 dB)
- **Unity Gain Frequency (UGF):** 4.872 MHz
- **−3 dB Bandwidth (BW):** 1.891 kHz
- **Gain-Bandwidth Product (GBW):** 5.051 MHz

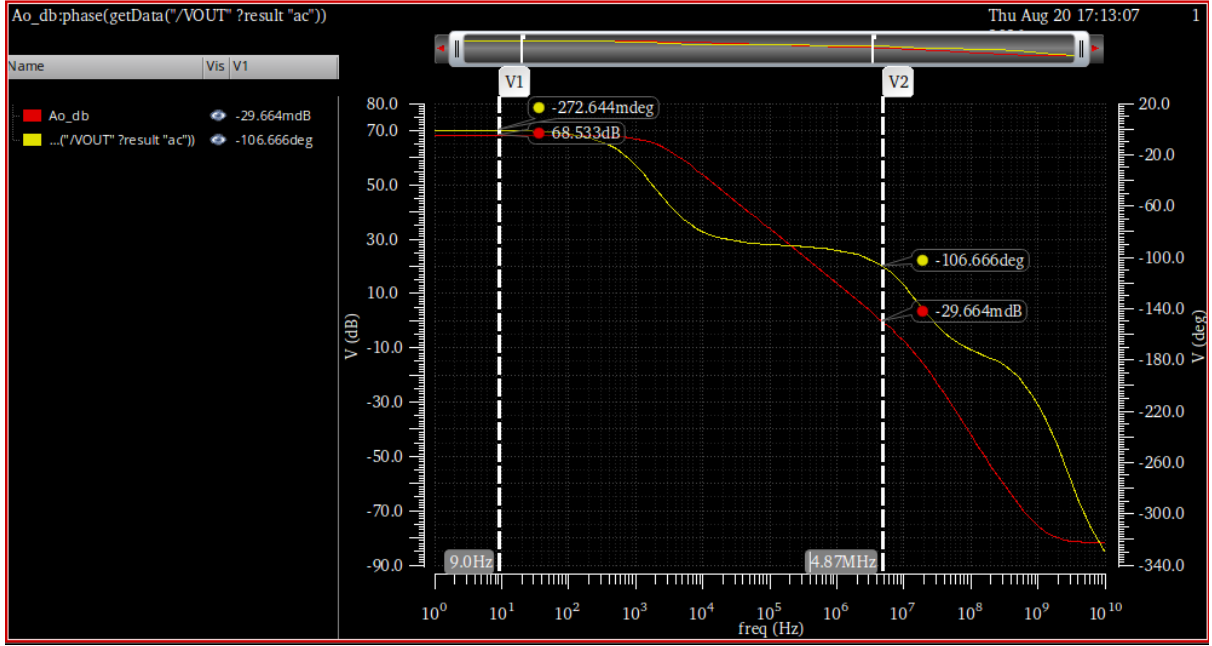


Figure 9: Bode plot of the open-loop differential response showing Gain (dB) and Phase (degrees).

2.3.1 Hand Analysis

1. Open-Loop DC Gain (A_v)

For the first stage (input pair M_8 , active load M_6):

$$r_{out1} = r_{o8} \parallel r_{o6} = \frac{(1.894 \text{ M}\Omega)(2.328 \text{ M}\Omega)}{1.894 \text{ M}\Omega + 2.328 \text{ M}\Omega} = 1.044 \text{ M}\Omega$$

$$A_{v1} = g_{m8} \cdot r_{out1} = (66.08 \mu\text{S})(1.044 \text{ M}\Omega) = 68.99 \text{ V/V}$$

For the second stage (CS driver M_7 , active load M_4):

$$r_{out2} = r_{o7} \parallel r_{o4} = \frac{(87.45 \text{ k}\Omega)(602.0 \text{ k}\Omega)}{87.45 \text{ k}\Omega + 602.0 \text{ k}\Omega} = 76.36 \text{ k}\Omega$$

$$A_{v2} = g_{m7} \cdot r_{out2} = (508.3 \mu\text{S})(76.36 \text{ k}\Omega) = 38.81 \text{ V/V}$$

The total DC gain is:

$$A_{v,\text{total}} = A_{v1} \cdot A_{v2} = (68.99)(38.81) = 2677 \text{ V/V}$$

$$A_{v,\text{dB}} = 20 \log_{10}(2677) = 68.55 \text{ dB}$$

Table 3: Comparison of Hand Analysis vs. Simulation Results

Parameter	(Hand Calc)	Simulated (Spectre)	Deviation
DC Gain (Linear)	2677 V/V	2671 V/V	0.22%
DC Gain (dB)	68.55 dB	68.53 dB	$\approx 0\%$

Design Note on Phase Margin and DC Centering: To improve the phase margin, the transconductance of the second stage (g_{m7}) was increased (By increasing w from 11.1u to 12.3u) to push the non-dominant pole to a higher frequency. However, this sizing adjustment shifted the DC operating point, disrupting the output voltage level. To resolve this and restore the DC output voltage to 900 mV ($V_{DD}/2$ for maximum symmetrical swing), the width of the first-stage active load ($M_{5,6}$) was re-adjusted to 5.4u and this enhance cmrr also. Additionally, throughout this process, the minimum input common-mode voltage ($V_{ICM,min}$) requirement was maintained due to the large margin during the initial design phase.(due to this gain is more than 2000 also)

2.4 Common-Mode Small Signal Characteristics

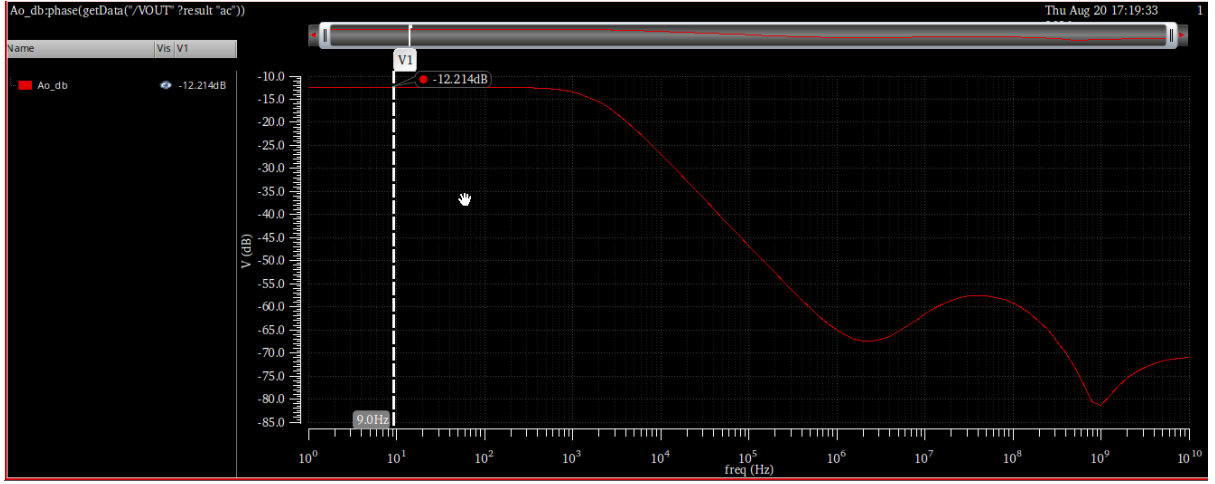


Figure 10: Bode plot of the open-loop common-mode response showing A_{cm} (dB).

Common-mode gain (A_{cm}) is observed to be -12.214 dB.

2.4.1 Hand Analysis

The magnitude of the first-stage common-mode gain is approximated by:

$$A_{cm1} \approx \frac{1}{2g_{m6}r_{o3}}$$

Using the extracted DC parameters:

$$A_{cm1} \approx \frac{1}{2(77.78 \mu S)(997.0 \text{ k}\Omega)} = \frac{1}{155.09} \approx 0.006448 \text{ V/V}$$

Because the common-mode error signal at the output of the first stage is fed directly into the second-stage common-source driver, it is amplified by the second stage's full differential gain (A_{v2}).

$$A_{cm,total} = A_{cm1} \cdot A_{v2} = (0.006448) \cdot (38.81) = 0.2502 \text{ V/V}$$

decibels:

$$A_{cm,dB} = 20 \log_{10}(0.2502) = -12.03 \text{ dB}$$

Parameter	(Hand Calc)	Simulated (Spectre)	Deviation
Common-Mode Gain (dB)	-12.03 dB	-12.21 dB	0.18 dB

2.5 Common-Mode Rejection Ratio (CMRR)

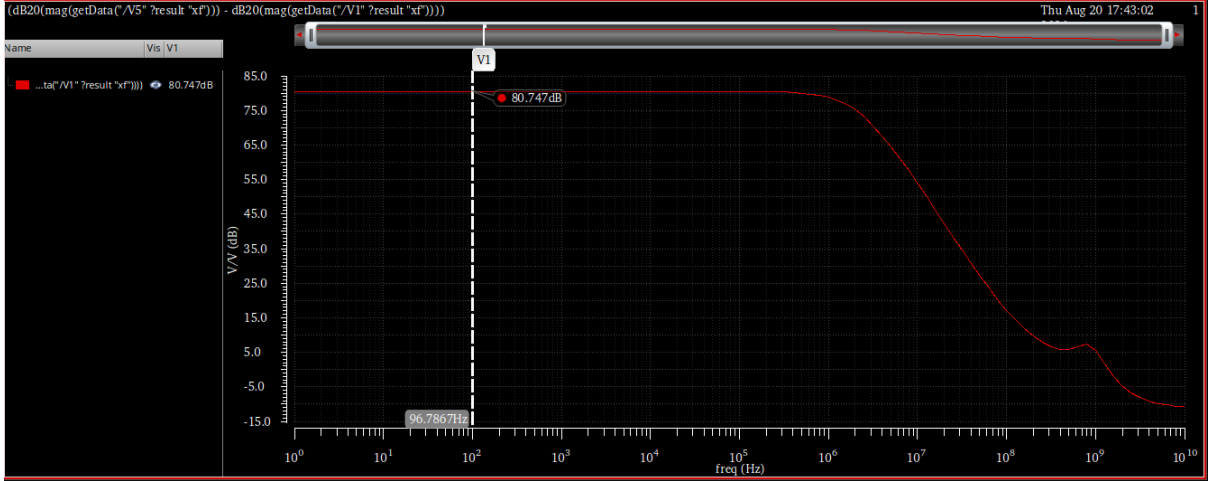


Figure 11: Simulation plot characterizing the input common-mode boundaries.

The Common-Mode Rejection Ratio (CMRR) is mathematically defined as the ratio of the differential-mode gain (A_{dm}) to the common-mode gain (A_{cm}). Expressed in decibels, this relationship simplifies to the algebraic difference between the two gains.

Using the simulated gain values:

$$\text{CMRR}_{\text{dB}} = 68.53 \text{ dB} - (-12.21 \text{ dB}) = 80.74 \text{ dB}$$

2.6 Differential Large Signal Characteristics

To evaluate the large-signal differential characteristics, a DC sweep is performed on the differential input voltage (V_{ID}) with the common-mode voltage set to $V_{ICM} = V_{DD}/3$.

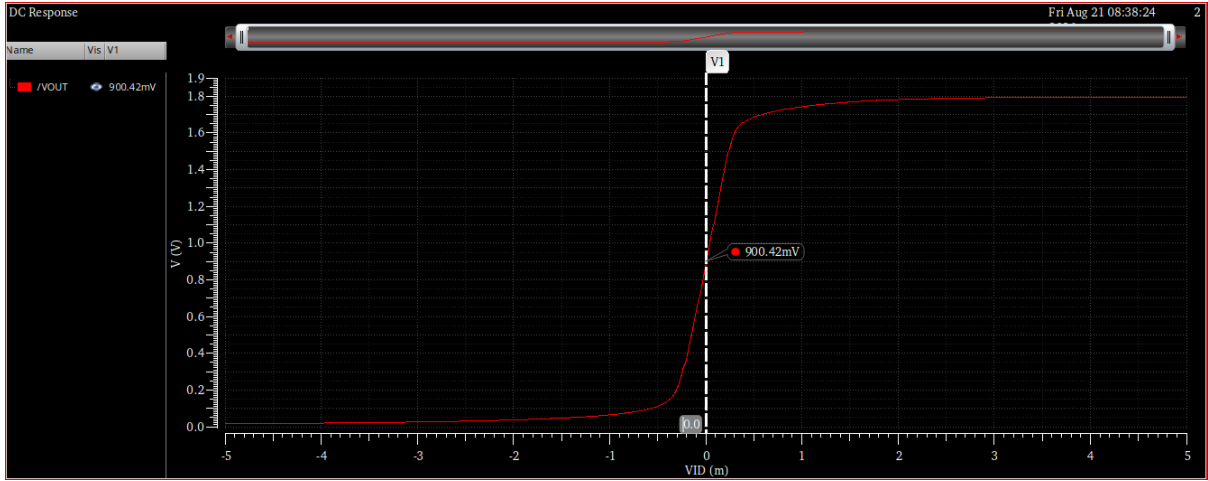


Figure 12: DC Response showing V_{OUT} versus V_{ID} .

From the DC sweep plot (Figure 12), the value of the output voltage at $V_{ID} = 0$ is 900.42 mV. This matches the resting DC output voltage of approximately 900 mV .

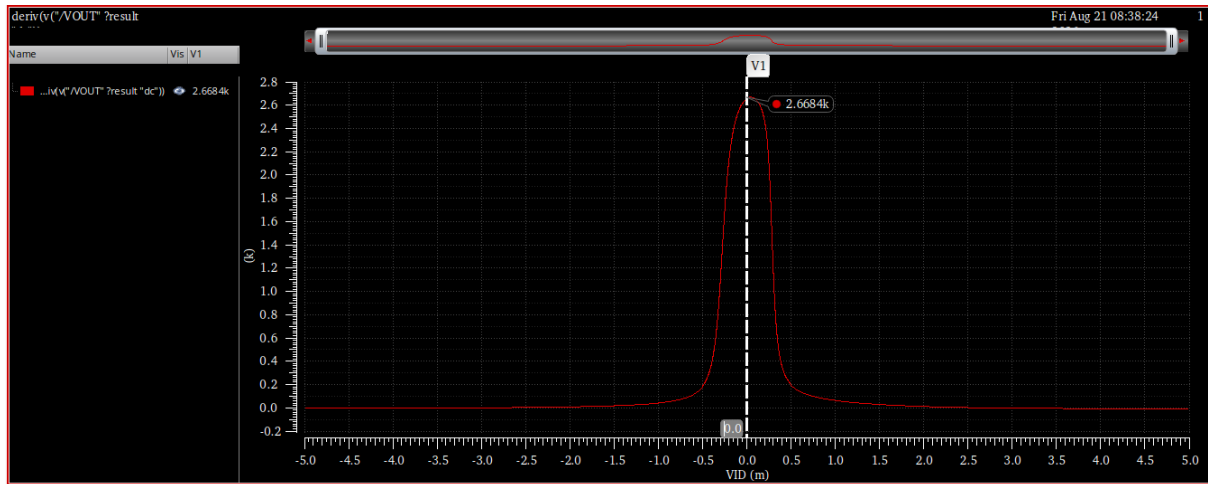


Figure 13: Derivative of the DC transfer characteristic (dV_{OUT}/dV_{ID}).

As shown in Figure 13, the peak value of the derivative is 2.6684 kV/V. The peak gain is slightly smaller than the A_{vd} value of 2.671 kV/V obtained from the AC analysis. This small difference due to the approximation used when calculating the derivative of the DC transfer characteristic from discrete DC sweep points, whereas the AC analysis calculates the small-signal gain directly from the linearized circuit .

2.7 Common-Mode Large Signal Characteristics (Region vs. V_{ICM})

To evaluate the boundaries of the Common-Mode Input Range (CMIR), a DC sweep of V_{ICM} from 0 V to 1.8 V was performed. The operating regions of the input differential pair (green trace) and the primary tail current source (yellow trace) were plotted. The simulator maps the operating regions numerically: 0 = cut-off, 1 = triode, 2 = saturation, and 3 = subthreshold.



Figure 14: Operating region of the input pair and tail current source versus V_{ICM} .

From Figure 14, the input pair remains in saturation (Region 2) all the way down to 0 V, taking advantage of the PMOS body effect. However, as V_{ICM} increases, the tail current source is compressed and eventually falls into the triode region (Region 1) at approximately 0.82 V. Therefore, the simulated valid common-mode input range is from 0 V to 0.82 V.

2.7.1 Hand Calculations

- **Minimum Common-Mode Voltage ($V_{ICM,min}$):**

The input PMOS transistors (M_8 , M_9) must stay in saturation. This requires $V_{DG} \leq |V_{thp}|$. The drain voltage of the input pair is fixed by the diode-connected active load (M_5), where $V_D = V_{GS5} \approx 0.731$ V.

$$V_{DG} = V_D - V_G = 0.731 \text{ V} - V_{ICM,min} \leq |V_{thp}|$$

$$V_{ICM,min} \geq 0.731 \text{ V} - |-0.779 \text{ V}| = -0.048 \text{ V}$$

- **Maximum Common-Mode Voltage ($V_{ICM,max}$):**

The tail current source (M_3) must remain in saturation, meaning $V_{SD3} \geq |V_{dsat3}|$.

$$V_{ICM,max} = V_{DD} - |V_{dsat3}| - V_{SG8}$$

Using the extracted values ($V_{DD} = 1.8$ V, $|V_{dsat3}| = 0.1385$ V, and $V_{SG8} = |V_{GS8}| = 0.862$ V):

$$V_{ICM,max} = 1.8 \text{ V} - 0.1385 \text{ V} - 0.862 \text{ V} \approx 0.80 \text{ V}$$

Table 5: Comparison of Hand Calculation vs. Simulation for CMIR

Parameter	Hand Calc	Simulated (Spectre)	Deviation
$V_{ICM,min}$	0 V	0 V	0%
$V_{ICM,max}$	0.80 V	0.82 V	5%

Design Note on Tail Current Source Sizing: To achieve the $V_{ICM,max}$ (CMIR-High) specification of 0.8 V (before that the tail drops at 790 mV), the g_m/I_D ratio of the tail current source (M_3) needed to be increased slightly. By performing a parameter sweep, allowing the CMIR-High requirement to be met without changing the other design specifications.

2.8 (Optional) Common-Mode Large Signal Characteristics (GBW vs. V_{ICM})

The valid CMIR is defined as the range over which the GBW is within 90% of its maximum value:.

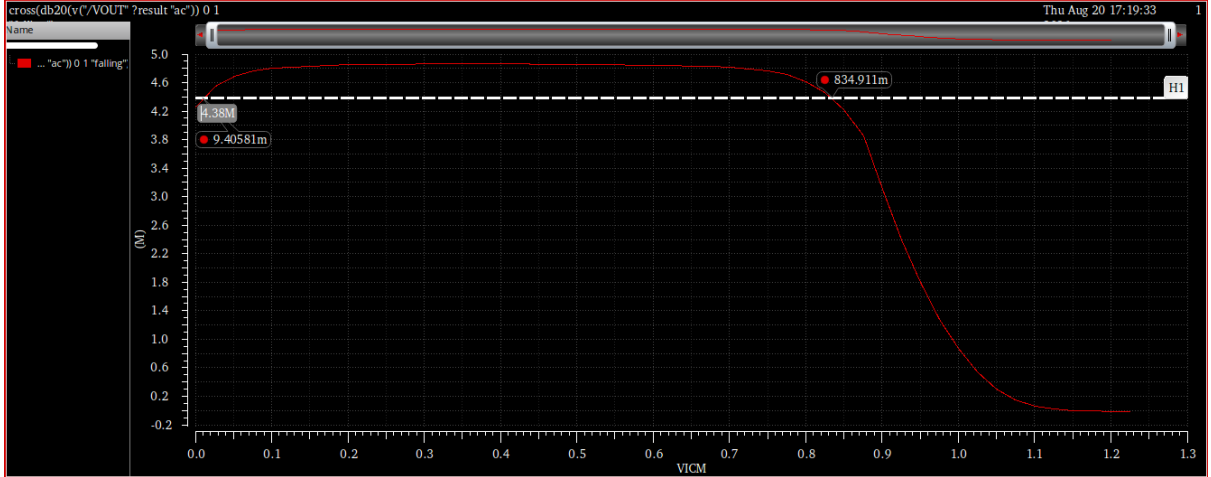


Figure 15: Simulated GBW (in MHz) versus input common-mode voltage (V_{ICM}).

From the simulation, the maximum GBW achieved by the amplifier is approximately 4.87 MHz. The 90% threshold is calculated as:

$$GBW_{90\%} = 0.9 \times 4.87 \text{ MHz} \approx 4.38 \text{ MHz}$$

As annotated by the horizontal marker in Figure 15, the GBW curve intersects this 4.38 MHz boundary at a lower bound of $V_{ICM} = 9.41 \text{ mV}$ and an upper bound of $V_{ICM} = 834.91 \text{ mV}$. Therefore, the amplifier maintains optimal performance over an input range from approximately 9.4 mV to 835 mV. This margin validates the target V_{ICMR} specification of 0.2 V to 0.8 V.

3 PART 4: Closed-Loop OTA Simulation

To evaluate the amplifier in a practical operational state, the OTA is connected in a unity-gain buffer feedback configuration. A zero-voltage source or current probe is inserted into the feedback loop to measure the loop gain and phase response.

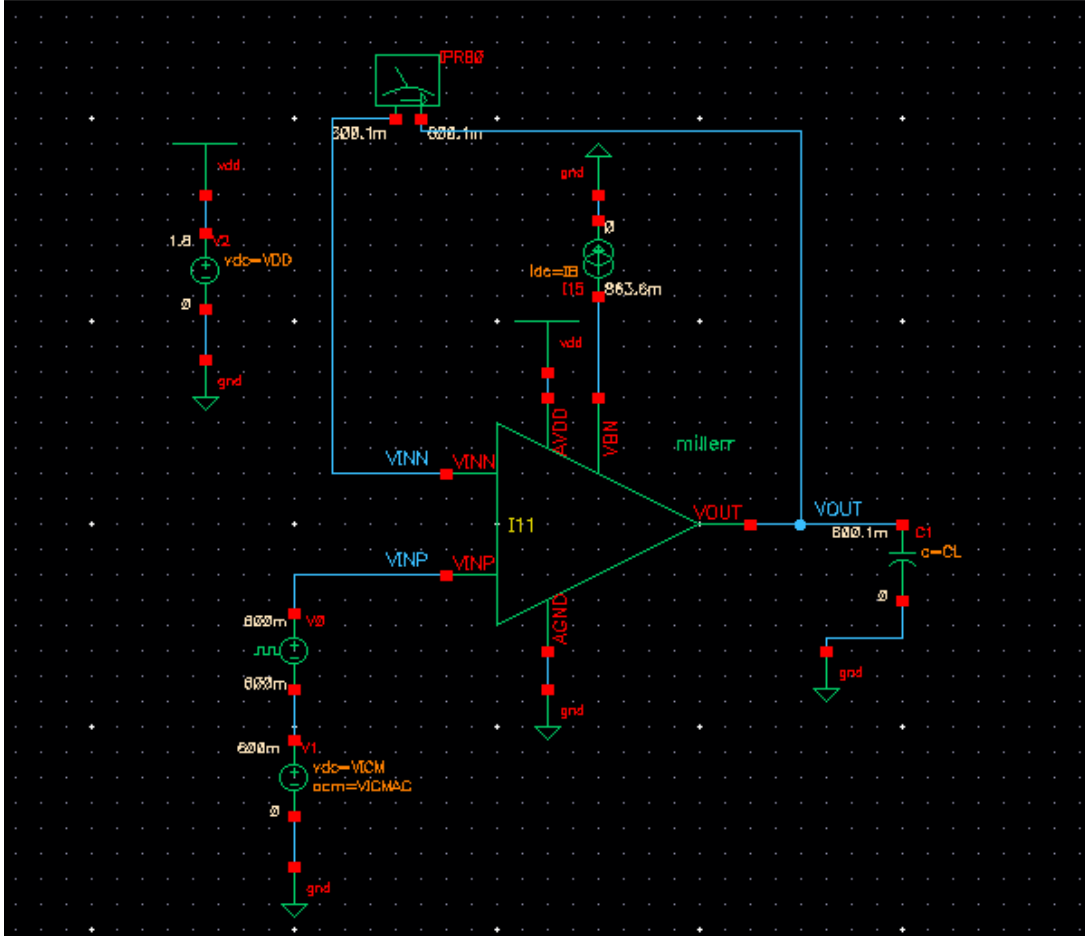


Figure 16: Testbench schematic of the OTA in a closed-loop unity-gain buffer configuration.

3.1 DC Operating Point

- **Are the DC voltages at the input terminals of the op-amp exactly equal? Why?**

The DC voltages at the two input terminals are approximately equal, but not exactly equal. This is due to the finite open-loop gain of the op-amp. Negative feedback forces the differential input voltage to a very small value, resulting in $V_+ \approx V_-$.

- **Is the DC voltage at the output of the first stage exactly equal to the value in the open-loop simulation? Why?**

No, it is approximately 739.2 mV. When you connect the amplifier in negative feedback, the loop automatically stabilizes the circuit. The feedback forces a small differential voltage at the input pins. This pushes the first stage slightly out of balance, raising its output node up to the 739.1 mV that M_7 requires to stabilize the output.

- **Is the current (and g_m) in the input pair exactly equal? Why?**

No, g_m and I_D are not exactly equal. Transistor mismatches and different vds cause a small difference between the two branch currents and g_m values.

The mismatch between the two input devices is proven by extracting their operating points, as shown below:

gm	66.08u
gmb	27.29u
gmbs	27.29u
gmoverid	13.69
ibe	731.3f
ibulk	209.7a
id	-4.826u

Figure 17: Operating point for inverting input (M_8).

gm	66.08u
gmb	27.3u
gmbs	27.3u
gmoverid	13.68
ibe	723.4f
ibulk	205.9a
id	-4.83u

Figure 18: Operating point for non-inverting input (M_9).

3.2 Loop Gain and Stability

A Stability (STB) analysis was performed to measure the loop gain magnitude and phase response across the frequency spectrum.

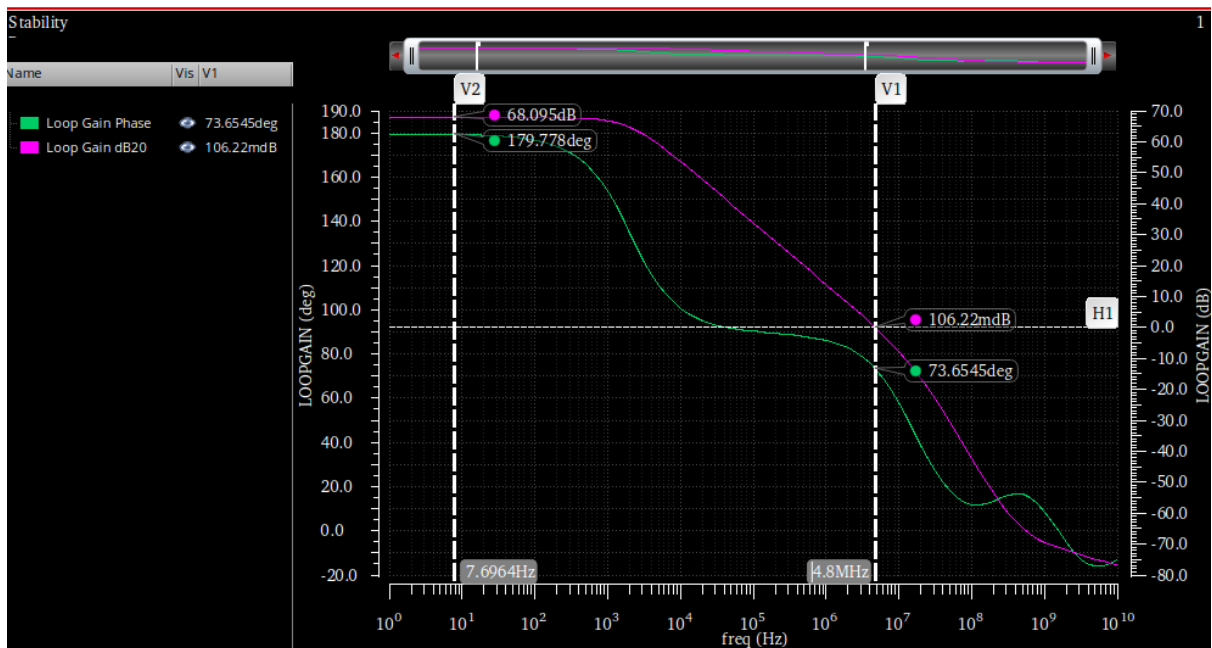


Figure 19: Bode plot of the closed-loop Phase (degrees) and Loop Gain (dB).

Test	Output	Nominal	Spec	Weight	Pass/Fail
miller:miller_tb:2	/VOUT				
miller:miller_tb:2	Loop Gain Phase				
miller:miller_tb:2	Loop Gain dB20				
miller:miller_tb:2	gainBwProd(getData("loopGain"...	5.047M			
miller:miller_tb:2	unityGainFreq(getData("loopGai...	4.867M			
miller:miller_tb:2	phaseMargin(getData("loopGai...	-106.5			

Figure 20: Simulated STB performance metrics (GBW, UGF, and Phase Margin).

3.2.1 Performance Comparison and Comments

- **Compare DC gain, f_u , and GBW with those obtained from open-loop simulation:**

The closed-loop loop gain is 68.09 dB, the Unity Gain Frequency (f_u) is 4.867 MHz, and the Gain-Bandwidth Product (GBW) is 5.047 MHz. The closed-loop DC gain is slightly reduced compared to the open-loop simulation (68.53 dB) due to the loading effect of the feedback. However, f_u and GBW remain almost identical to the open-loop results, as expected in a unity-gain configuration.

- **Report PM and compare with hand calculations:**

The STB simulation reports a Phase Margin (PM) of 73.65°. This matches closely with our hand calculations and exceeds the $\geq 70^\circ$ target design specification.

3.2.2 Simple Hand Calculations

Phase Margin are estimated as:

$$f_u \approx \frac{g_{m1}}{2\pi C_c} = \frac{66.08 \mu\text{S}}{2\pi(2 \text{ pF})} \approx 5.26 \text{ MHz}$$

$$f_{p2} \approx \frac{g_{m7}}{2\pi C_L} = \frac{508.3 \mu\text{S}}{2\pi(5 \text{ pF})} \approx 16.18 \text{ MHz}$$

$$\text{PM} \approx 90^\circ - \arctan\left(\frac{f_u}{f_{p2}}\right) = 90^\circ - \arctan\left(\frac{5.26}{16.18}\right) \approx 72^\circ$$

Table 6: Comparison of Hand Calculations vs. Open-Loop and Closed-Loop Simulations

Parameter	Hand Calc	Open-Loop Sim	Closed-Loop Sim (STB)
DC Gain (dB)	68.55 dB	68.53 dB	68.09 dB
Unity Gain Frequency (f_u)	5.26 MHz	4.87 MHz	4.86 MHz
Phase Margin (PM)	72°	73.33°	73.65°

3.3 Slew Rate

Initially, the simulated slew rate was measured at 4.383 V/ μs , which failed to meet the target specification of $\geq 5 \text{ V}/\mu\text{s}$.

Test	Output	Nominal	Spec	Weight	Pass/Fail
millerr_tb:2	/VOUT				
millerr_tb:2	/VINP				
millerr_tb:2	ymax(deriv(v("/VOUT" ?result "tr...	4.383M			

Figure 21: Initial failing simulated slew rate of 4.383 V/ μs .

To resolve this issue, the Miller compensation capacitor (C_c) was decreased from 2 pF to 1.7 pF. This modification improved the slew rate while ensuring the phase margin specification remained valid. Because of the reduced compensation capacitance, the

Unity Gain Frequency (UGF) increased to 5.68 MHz. Consequently, the phase margin decreased to approximately 70° ($180^\circ - 109.334^\circ$), which is acceptable and meets the stability requirements.

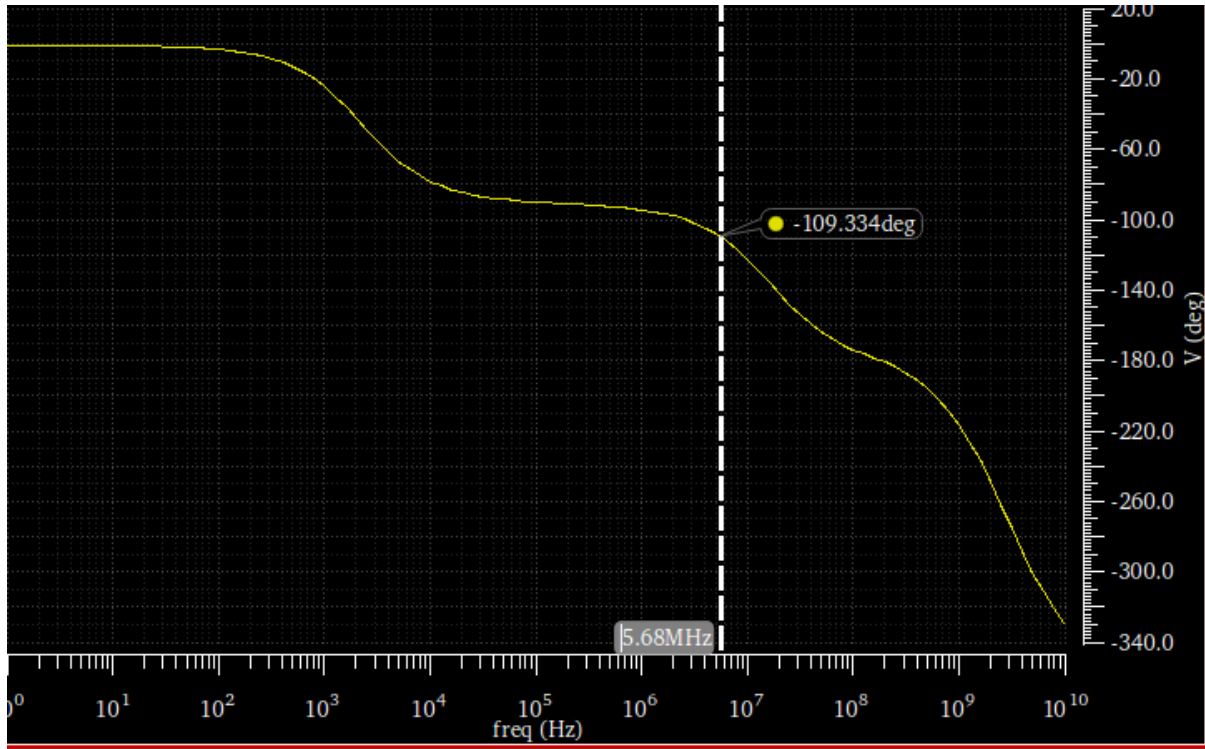


Figure 22: Open-loop phase response showing an approximate 70° phase margin at the new UGF of 5.68 MHz.

The new simulated slew rate reached $5.02 \text{ V}/\mu\text{s}$, achieving the design specification.

Test	Output	Nominal	Spec	Weight	Pass/Fail
miller:millerr_tb:2	/VOUT				
miller:millerr_tb:2	/VINP				
miller:millerr_tb:2	ymax(deriv(v("/VOUT" ?result "tr...	5.02M			

Figure 23: Final simulated slew rate achieving $5.02 \text{ V}/\mu\text{s}$.

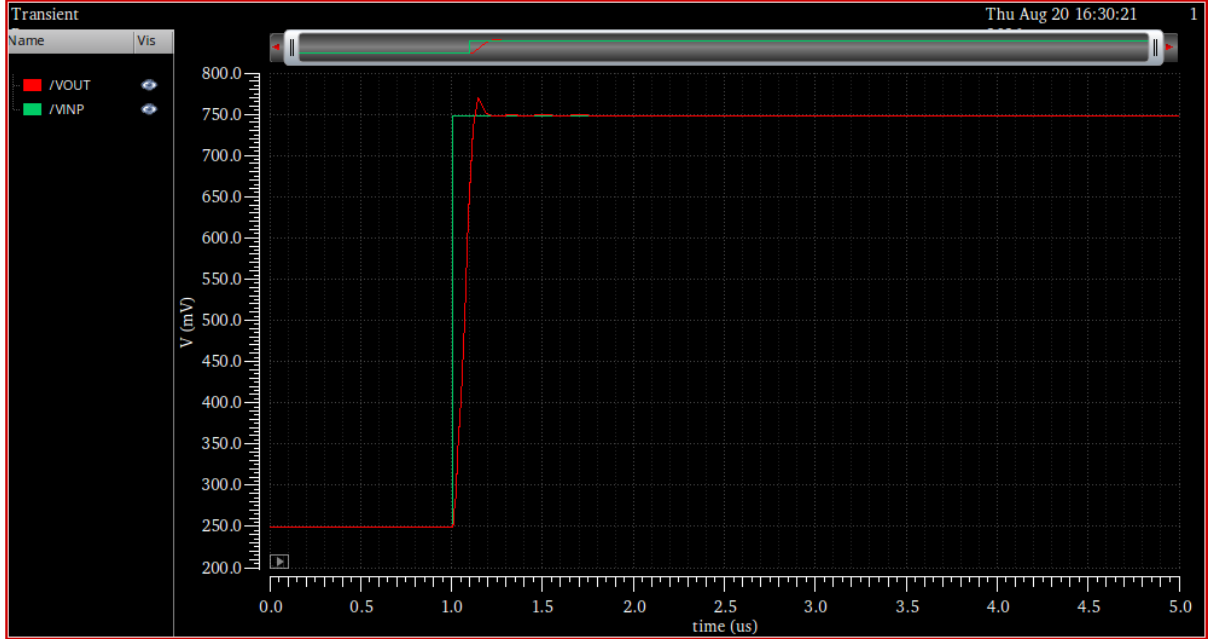


Figure 24: Transient response showing V_{IN} and V_{OUT} overlaid during a large-signal step.

3.3.1 Hand Calculation

The internal slew rate of the OTA the total tail bias current (I_B) available to charge and discharge the compensation capacitor (C_c). Using the simulated tail current of $I_B = 9.64 \mu\text{A}$ and the updated compensation capacitance of $C_c = 1.7 \text{ pF}$, the expected slew rate is calculated as:

$$SR = \frac{I_B}{C_c}$$

$$SR = \frac{9.64 \mu\text{A}}{1.7 \text{ pF}} \approx 5.67 \text{ V}/\mu\text{s}$$

Table 7: Comparison of Hand Calculation vs. Simulation for Slew Rate

Parameter	Hand Calc	Simulated (Spectre)	Deviation
Slew Rate (SR)	$5.67 \text{ V}/\mu\text{s}$	$5.02 \text{ V}/\mu\text{s}$	11.46%

3.4 Settling Time and Small-Signal Rise Time



Figure 25: Small-signal transient response showing the output rise time and settling behavior.

miller:millerr_tb:2	riseTime(VT("/VOUT") 0.6 nil 0.6...	43.87n			
---------------------	-------------------------------------	--------	--	--	--

Figure 26: Extracted small-signal rise time measurement (43.87 ns).

3.4.1 Hand Calculation

For a single-pole (first-order) system approximation, the closed-loop time constant (τ) is inversely proportional to the unity-gain frequency ω_u . The rise time (t_{rise}) for a first-order system is calculated as:

$$t_{rise} = 2.2\tau$$

$$\tau = \frac{1}{\omega_u}$$

Using the updated Unity Gain Frequency ($f_u = 5.68$ MHz) established in the previous section:

$$\omega_u = 2\pi f_u = 2\pi (5.68 \times 10^6) \approx 35.68 \times 10^6 \text{ rad/s}$$

$$\tau = \frac{1}{35.68 \times 10^6} \approx 28.02 \text{ ns}$$

$$t_{rise} = 2.2 \times 28.02 \text{ ns} \approx 61.64 \text{ ns}$$

Table 8: Comparison of Hand Calculation vs. Simulation for Small-Signal Rise Time

Parameter	Hand Calc (1st Order)	Simulated (Spectre)	Deviation
Rise Time (t_{rise})	61.64 ns	43.87 ns	28.8%

3.4.2 Comments on the Results

- **Deviation in Rise Time:** The simulated rise time (43.87 ns) is faster than the hand-calculated value (61.64 ns). Because of the $t_{rise} = 2.2\tau$ formula is based on a first-order . In reality, the two-stage OTA is a second-order system. Because the phase margin was reduced to approximately 70° , the system is slightly underdamped. Second-order underdamped systems exhibit faster rise times than first-order systems.
- **Observation of Ringing:** A slight ringing (overshoot) is visible at the peak of the step response in Figure 25. This is expected. A system with a phase margin of less than 90° contains complex conjugate poles. In addition, a phase margin of $\approx 70^\circ$ corresponds to a damping factor (ζ) of 0.6 to 0.7, which produces a small overshoot before the output settles to its final value.

4 PART 5: DC Closed-Loop AC Open-Loop OTA Simulation

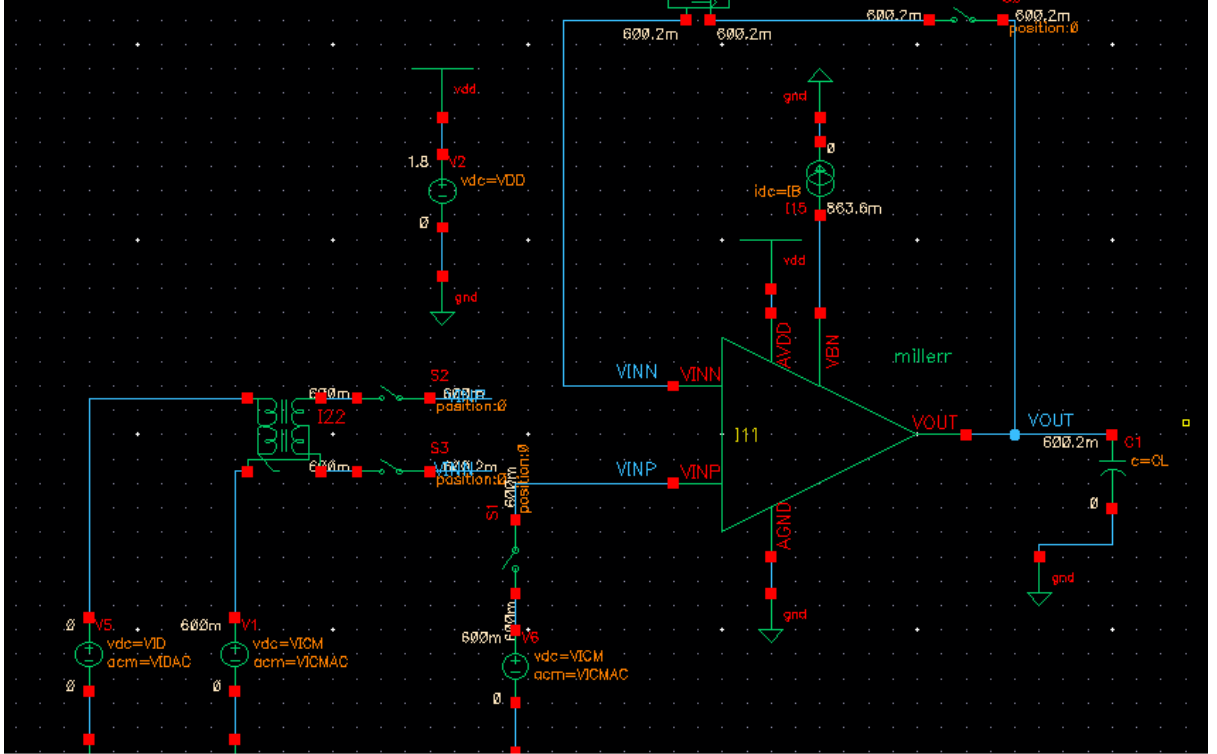


Figure 27: Testbench schematic utilizing ideal switches to separate the DC biasing loop from the AC open-loop stimulus.

Circuit Configuration and Behavior

- **The Feedback Loop:** By setting switch S_0 to be closed in DC, the output node (V_{OUT}) is connected directly back to the inverting input (V_{INN}). This effectively puts the amplifier in a standard unity-gain buffer configuration for the DC solution.

- **The Reference Voltage:** By setting switch S_1 to be closed in DC, the non-inverting input (V_{INP}) is tied directly to the common-mode DC voltage source (V_{ICM}).
- **Operating Point Stabilization:** Because the OTA is placed in a negative feedback loop for the DC calculation, it will drive its output to force the difference between its two input to zero. Since the positive pin is pinned to $V_{ICM} = 600\text{ mV}$, the amplifier should push V_{OUT} to 600 mV .

5 PART 6: Tracking Lead Compensation

Using an ideal fixed resistor for lead compensation is impractical in physical layout. Furthermore, as temperature fluctuates, the transconductance of the second-stage driving transistor (g_{m7}) drifts. If R_Z remains fixed while g_{m7} changes, the LHP zero will no longer cancel the non-dominant pole, degrading the phase margin over varying PVT conditions.

Following the Johns and Martin tracking lead compensation methodology, the ideal resistor is replaced with an NMOS transistor operating in the deep triode region. To ensure the zero tracks the non-dominant pole, the resistance R_Z must track the inverse transconductance of the second stage ($1/g_{m7}$). Therefore, the design goal is to force the product $R_Z g_{m7} = 1$.

To achieve this, a bias network is constructed to generate a tracking gate voltage for the triode resistor. By matching the effective gate-source voltages of the bias network to the second stage, the product $R_Z g_{m7}$ becomes dependent on transistor geometry:

$$R_Z g_{m7} = \frac{(W/L)_{M7}}{(W/L)_{triode}} \sqrt{\frac{(W/L)_{top_diode}}{(W/L)_{bottom_diode}}}$$

Using this equation, the compensation network is sized as follows:

Step 1: Sizing the Bias Network

A $10\text{ }\mu\text{A}$ bias branch is created via a PMOS current mirror. To match the effective voltages, the W/L ratio must equal the current ratio between the second stage (M_7) and the bias branch:

$$\frac{I_{D(M7)}}{I_{D(bias)}} = \frac{40\text{ }\mu\text{A}}{10\text{ }\mu\text{A}} = 4 \implies (W/L)_{bottom_diode} = \frac{1}{4}(W/L)_{M7}$$

Given M_7 has a width of $12.3\text{ }\mu\text{m}$ and a length of 320 nm , the bottom diode is sized to $W = 3.075\text{ }\mu\text{m}$ with $L = 320\text{ nm}$. To form the tracking stack, the top diode is sized identically to the bottom diode ($W = 3.075\text{ }\mu\text{m}$).

Step 2: Sizing the Triode Resistor

Because the top and bottom diodes are geometrically identical, the square root term in the master equation cancels out ($\sqrt{1} = 1$). To place the zero at infinity ($R_Z = 1/g_{m7}$), the remaining geometry ratio must equal 1:

$$1 = \frac{(W/L)_{M7}}{(W/L)_{triode}} \times 1 \implies (W/L)_{triode} = (W/L)_{M7}$$

Thus, the triode resistor is sized identically to the second-stage driver M_7 : $W = 12.3\text{ }\mu\text{m}$ and $L = 320\text{ nm}$.

Complete Schematic with Tracking Lead Compensation

The final schematic incorporating the dynamically biased tracking resistor is shown below.

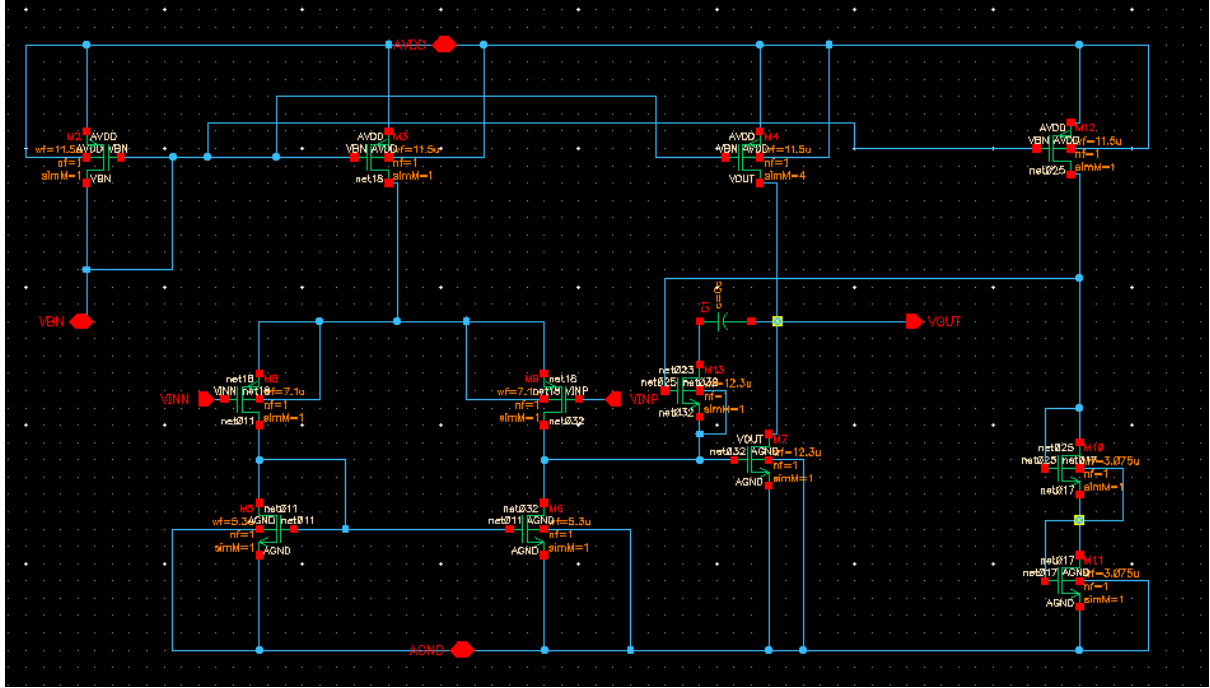


Figure 28: lead compensation within the Two-Stage Miller OTA schematic. .

Nominal Verification

Running the STB analysis at nominal conditions confirms the tracking circuit replicates the ideal resistor's behavior, locking in the target phase margin.

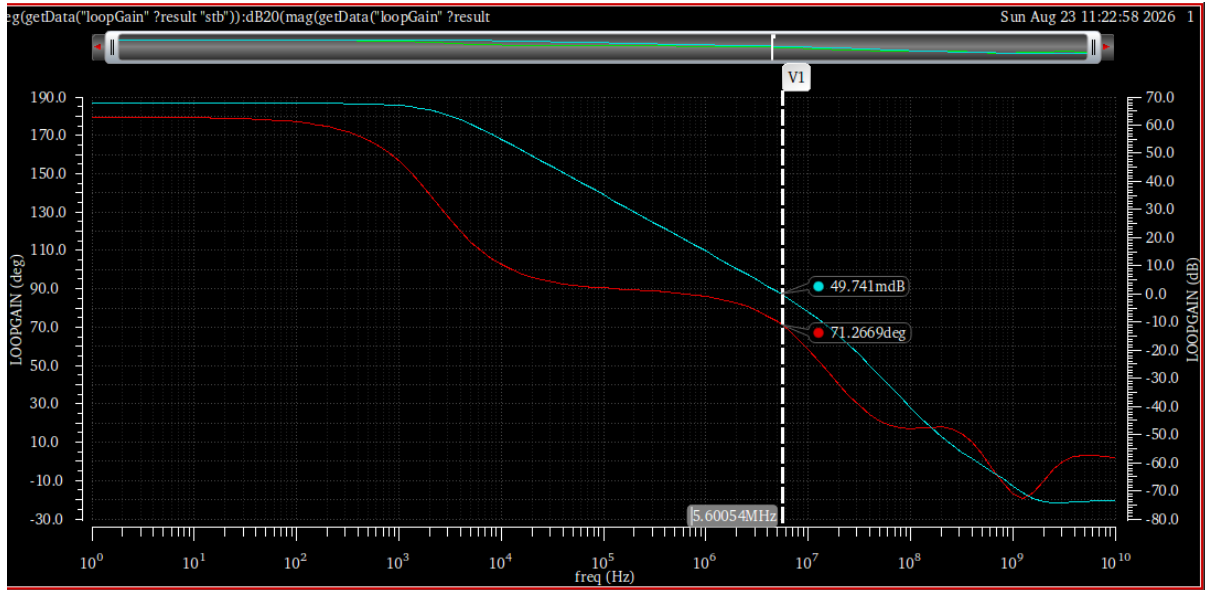


Figure 29: STB loop gain and phase response with the tracking triode resistor.

miller:millerr_tb:2	gainBwProd(getData("loopGain"...	5.943M			
miller:millerr_tb:2	phaseMargin(getData("loopGai...	-108.7			
miller:millerr_tb:2	unityGainFreq(getData("loopGai...	5.671M			

Figure 30: Extracted STB parameters showing a 5.67 MHz UGF and -108.7° phase at new value of Cc=(1.7p).

As shown in the simulation results, the Unity Gain Frequency (UGF) is maintained at 5.67 MHz, and the Phase Margin is locked at 71.3° (calculated as 180° - 108.7°). This confirms that the dynamically biased NMOS transistor tracks the transconductance and nullifies the right-half-plane zero without degrading the established open-loop AC characteristics.

6 Conclusion and Design Summary

6.1 Area and Current Consumption

Table 9: Main OTA Active Transistor Area Calculation

Device	Width (W)	Length (L)	Multiplier (M)	Total Area ($W \times L \times M$)
M_8 (Inverting Input)	$7.1 \mu\text{m}$	$0.43 \mu\text{m}$	1	$3.053 \mu\text{m}^2$
M_9 (Non-Inverting Input)	$7.1 \mu\text{m}$	$0.43 \mu\text{m}$	1	$3.053 \mu\text{m}^2$
M_5 (Stage 1 Load Ref)	$5.4 \mu\text{m}$	$0.43 \mu\text{m}$	1	$2.322 \mu\text{m}^2$
M_6 (Stage 1 Load Mirror)	$5.4 \mu\text{m}$	$0.43 \mu\text{m}$	1	$2.322 \mu\text{m}^2$
M_3 (Tail Current)	$11.2 \mu\text{m}$	$0.60 \mu\text{m}$	1	$6.720 \mu\text{m}^2$
M_2 (Bias Mirror)	$11.2 \mu\text{m}$	$0.60 \mu\text{m}$	1	$6.720 \mu\text{m}^2$
M_7 (Stage 2 Driver)	$12.3 \mu\text{m}$	$0.32 \mu\text{m}$	1	$3.936 \mu\text{m}^2$
M_4 (Stage 2 Load)	$11.2 \mu\text{m}$	$0.60 \mu\text{m}$	4	$26.880 \mu\text{m}^2$
Main OTA Transistor Area				$55.006 \mu\text{m}^2$

Table 10: Tracking Lead Compensation Active Transistor Area

Device	Width (W)	Length (L)	Multiplier (M)	Total Area ($W \times L \times M$)
PMOS Bias Mirror	$11.2 \mu\text{m}$	$0.60 \mu\text{m}$	1	$6.720 \mu\text{m}^2$
M_9 (Top Diode)	$3.075 \mu\text{m}$	$0.32 \mu\text{m}$	1	$0.984 \mu\text{m}^2$
M_{10} (Bottom Diode)	$3.075 \mu\text{m}$	$0.32 \mu\text{m}$	1	$0.984 \mu\text{m}^2$
M_{11} (Triode Resistor)	$12.3 \mu\text{m}$	$0.32 \mu\text{m}$	1	$3.936 \mu\text{m}^2$
Compensation Transistor Area				$12.624 \mu\text{m}^2$
Total Combined Area				$67.630 \mu\text{m}^2$

The total internal current is the sum of the first-stage tail current and the second-stage driving current:

$$I_{total} = I_{M3} + I_{M7}$$

$$I_{total} \approx 9.66 \mu\text{A} + 39.94 \mu\text{A} = 49.60 \mu\text{A}$$

This value satisfies the design requirement of $\leq 60 \mu\text{A}$ (It is also noted that the current mirror exhibits a mismatch of 2.4%) (and I have ability to make the circuit of RZ without break the spec of $I \leq 60 \mu\text{A}$).

6.2 Final Performance vs. Specifications

The following table summarizes the simulated performance of the Two-Stage Miller OTA against the target design specifications for the $0.18 \mu\text{m}$ technology node.

Table 11: Final Design Performance vs. Target Specifications

Parameter	Target Specification	Simulated Result
Supply Voltage	1.8 V	1.8 V
Static Gain Error	$\leq 0.05\%$ ($A_{ol} \geq 66.02 \text{ dB}$)	0.037% ($A_{ol} = 68.53 \text{ dB}$)
CMRR @ DC	$\geq 74 \text{ dB}$	80.74 dB
Phase Margin (PM)	$\geq 70^\circ$	71.3° (at $C_c = 1.7 \text{ pF}$)
OTA Current Consumption	$\leq 60 \mu\text{A}$	$49.60 \mu\text{A}$
CMIR - Low	$\leq 0.2 \text{ V}$	0.009 V
CMIR - High	$\geq 0.8 \text{ V}$	0.835 V
Buffer Closed-Loop Rise Time	$\leq 70 \text{ ns}$	43.87 ns
Slew Rate (SR)	$\geq 5 \text{ V}/\mu\text{s}$	$5.02 \text{ V}/\mu\text{s}$

The design meets all primary DC, small-signal, and large-signal transient specifications while maintaining stability and an efficient current footprint.